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COMPLEX SPIKES IN PURKINJE CELLS OF THE PARAVERMAL PART OF THE ANTERIOR LOBE OF THE CAT CEREBELLUM DURING LOCOMOTION

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SUMMARY

1. The temporal pattern of the discharge of complex spikes by Purkinje cells in the paravermal cortex of the cerebellar lobule V b/c has been examined during locomotion in awake cats.

2. The peripheral receptive fields of 138 Purkinje cells were examined using light tactile stimulation. In 91% of these cells, complex spikes were evoked by stimuli applied to the ipsilateral forelimb and of eighty-eight cells examined in most detail, 76% had receptive fields including the paw or wrist. Sixty-six per cent had receptive fields restricted to the paw and/or wrist.

3. Complex spikes were not discharged at rigidly fixed times during the step cycle in any of sixty-nine Purkinje cells which were recorded during locomotion on a moving belt.

4. When the discharges were averaged over many steps the probability of occurrence of complex spikes showed small fluctuations during the course of the step cycle, but these fluctuations were shown not to be statistically significantly different from those which could arise by chance.

5. These findings are inconsistent with previous suggestions (e.g. Armstrong, 1974; Rushmer, Roberts & Augter, 1976) that, during locomotion, the climbing fibres act to signal the occurrence of specific peripheral events, such as foot touch-down or lift-off.

INTRODUCTION

The discharges of simple spikes by the Purkinje cells of the cerebellar anterior lobe are strongly frequency modulated during locomotion (Orlovsky, 1970; Armstrong & Edgley, 1984*b*; Edgley & Lidierth, 1988). Such simple spikes result from the activation of Purkinje cells by mossy fibres via the intervening granule cells while a

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second, much less frequent type of action potential, the complex spike, results from direct activation of the Purkinje cells by the climbing fibres of the olivo-cerebellar projection. This report is concerned with the discharges of these complex spikes during locomotion in the cat.

Complex spikes are often readily discharged in response to mechanical stimulation of the periphery but, as they generally occur singly in response to such stimuli, individual cells appear to be ill-suited to the task of conveying graded information about stimulus magnitude on a trial-by-trial basis (Rushmer *et al.* 1976; cf. Kolb & Rubia, 1980). This has led to suggestions that they may instead act to indicate the occurrence of specific peripheral events during movement, and with particular reference to locomotion, that they may indicate the occurrence of foot touch-down or lift-off (Armstrong, 1974; Rushmer *et al.* 1976); however, the experimental evidence relevant to this hypothesis is equivocal. Thus, several studies have failed to demonstrate activity patterns that are consistent with such an event-detector hypothesis, either in the discharges of olive cells in the walking decerebrate cat (Boylls, 1980; eighteen cells in the caudal medial accessory olive which projects to the vermal a zone; see Brodal & Kawamura, 1980; Voogd & Bigare, 1981, for cortical zonation) or in the complex spike discharges of Purkinje cells in the awake cat walking on a moving belt (Andersson & Armstrong, 1987; nineteen cells in the vermal b zone), or on a horizontal ladder (Andersson & Armstrong, 1987; four cells in the paravermal cortex). Nevertheless, the matter warrants further attention because other studies have demonstrated step-related modulation in the discharges of complex spikes. Thus, in contrast to Andersson & Armstrong (1987), Udo, Matsukawa, Kamei, Minoda & Oda (1981) have observed a clear step-cycle-related modulation in the discharge probability of complex spikes such that the discharge rate was markedly increased during the E_1 phase of the step cycle of the ipsilateral forelimb in each of ten Purkinje cells recorded from the vermal b zone of decerebrate cats during quadrupedal stepping. In addition, a recent report by Kim, Wang & Ebner (1987) has demonstrated clear modulation in the discharges of complex spikes in thirty-six of sixty-seven Purkinje cells recorded in the paravermal cortex of lobule V (i.e. in the same cortical area that has been examined here) in a decerebrate cat preparation similar to that of Udo *et al.* (1981). The majority of these cells exhibited increased complex spike activity during the periods of transition between the swing and stance phases of the step cycle (i.e. at around the times of foot-fall and foot-lift).

The present paper describes the complex spike discharges of Purkinje cells in the paravermal cortex of lobule V c. As the receptive fields of these cells are concentrated on the forepaw and as they respond to light tactile stimuli, they are more likely to behave in the manner suggested by Armstrong (1974) and Rushmer *et al.* (1976) than are the Purkinje cells of the vermis which have more diffuse receptive fields (generally involving proximal rather than distal parts of the limbs; Armstrong & Andersson, 1987) and discharges which are less strongly related to peripheral events (Andersson & Sjölund, 1978; Sjölund, 1978). Furthermore, as this is the same area of cortex in which Kim *et al.* (1987) demonstrated step-related modulation in the discharges of complex spikes, the present data may be directly compared with those in the most extensive study to have demonstrated such modulation in decerebrate preparations.

Some preliminary data have been reported elsewhere (Armstrong, Campbell, Edgley, Schild & Trott, 1981).

METHODS

Extracellular recordings were made from the cerebellar cortex in awake cats using glass-insulated tungsten microelectrodes. Simultaneous recordings were made of the electromyographic activity of selected limb muscles which always included the triceps brachii muscle in the ipsilateral forelimb. Full details of the techniques of animal training, preparative surgery, receptive field determination and data recording have been provided previously (Armstrong & Drew, 1984; Armstrong & Edgley, 1984*a*, *b*).

Data analysis

Most Purkinje cells discharged both complex spikes and simple spikes which could be distinguished by their shape. Complex spikes consisted of an initial spike, which was generally similar to the simple spikes but often of larger amplitude and of longer duration. This initial spike was followed by one or more secondary spikes or wavelets which were not present following a simple spike (see Fig. 1). Only those Purkinje cells in which complex spikes were reliably discriminable from simple spikes have been included in the present analysis in order to obviate any possibility that simple spikes might have been erroneously identified as complex spikes. In addition, some cells have been included in which the discharge consisted only of complex spikes (cf. Thach, 1970; Armstrong & Rawson, 1979; Kolb & Rubia, 1980).

Recordings were analysed either manually from sections of data filmed with a Grass C4N camera or by computer following electronic discrimination of the complex spikes using a voltage–time window discriminator. The timings of the discharges of complex spikes during the step cycle were measured relative to the onset of electromyographic activity in the ipsilateral triceps brachii muscle (an elbow extensor) and post-event time histograms with a bin width of 100 ms were then constructed for each data sample. For those cells analysed by computer ($n = 41$), normalized histograms with a bin width equal to 1/10th the average step cycle duration were also constructed. The 'modulation depth' of each cell was taken as the firing rate for the bin of the normalized histogram in which the cell was most active minus the rate in the bin in which the cell was least active.

RESULTS

General

Recordings were made from several hundred Purkinje cells in the paravermal cortex of lobule V b/c in nineteen cats. The simple spike discharges of these cells are described elsewhere (Armstrong & Edgley, 1984*b*; Edgley & Lidierth, 1988). Only those cells with consistently discriminable complex spikes were included in the present analysis.

Tactile receptive fields

The peripheral receptive fields in which light manual tapping or gentle squeezing could evoke the discharge of a complex spike were determined in 138 Purkinje cells recorded from eight of the cats in which recording chambers were mounted above the lateral part of the paravermal cortex, corresponding to Oscarsson's c_2 and c_3 zones (e.g. Oscarsson, 1979).

In 126 cells (91%) the complex spike receptive field included the ipsilateral forelimb, and sixteen of ninety-three tested cells (17%) received input from both forelimbs. The receptive fields of those cells not activated from the forelimb included the hindlimb(s), face and thorax.

The receptive fields were examined in more detail in eighty-six cells. For the majority of these the receptive fields on the ipsilateral forelimb were restricted to the paw ($n = 15/86$; 17%), the wrist ($n = 8/86$; 9%) or the paw plus wrist ($n = 34/86$; 40%). In addition, eight cells (9%) had receptive fields restricted to the elbow and four (5%) to the upper arm and shoulder regions. In the remaining seventeen (20%) cells the receptive fields were more diffuse and covered larger areas of the limb, but in eight of these cells this diffuse area included the paw or wrist.

For those cells with bilateral input, the receptive fields on the contralateral forelimb covered areas that were homologous with those on the ipsilateral limb and were therefore on the distal parts of the limb in most cases. In each of fifty-nine cells for which receptive fields were determined for both simple and complex spikes the former were evoked from the same areas in which stimulation evoked a complex spike. However, in some cells in which complex spikes were evoked exclusively by ipsilateral stimuli, simple spikes could be evoked by stimulation of either forelimb ($n = 6/26$; 23%).

In cats (eleven out of nineteen) in which recordings were made in the medial part of the paravermal cortex (i.e. from Oscarsson's c_1 zone and the medial part of the c_2 zone; see Oscarsson, 1979), the receptive fields for simple spikes were similar to those in the lateral part of the cortex. Unfortunately, recording stability in these cats was not as good as among those with chambers over the lateral paravermal cortex and the complex spike receptive fields were therefore not determined because of the time-consuming nature of this procedure.

Discharges of complex spikes during locomotion

The complex spikes of sixty-nine Purkinje cells were recorded over a sequence of forty or more succesive, regular steps on a belt moving at 0·5–0·7 m/s. Only periods of locomotion in which the coefficient of variation of the step cycle duration was less than 10% of the mean duration were included in the present sample. Unfortunately, it was rarely possible to maintain stable recordings in which complex spikes were reliably discriminable for a sufficient time both to adequately determine a cell's receptive field and to record its step-related activity for long enough to make analysis worthwhile. The sample presented here is therefore, by and large, comprised of different cells to those used in the receptive field analysis above. However, the two samples were drawn from the same cats and as the cortical areas available for recording in each cat were restricted to a cylinder of *ca.* 2 mm in diameter, the distribution of receptive fields in the two samples should be essentially identical.

Twenty-eight of these cells were recorded from the medial part, and forty-one from the lateral part, of the paravermal cortex. Examples of the discharges of three cells are shown in Fig. 1.

The mean rate at which complex spikes were discharged during locomotion was $1·5 \pm 0·3$ impulses/s (mean ± S.D.; $n = 69$) which is the same as the mean rate observed in a sample of thirty-three cells recorded in the resting cat. Thirty-one cells were recorded both during rest and during locomotion and in these cells there was a statistically significant tendency for the mean rate to increase during locomotion (Wilcoxon signed-ranks test, $P < 0·001$; Siegel, 1956) although the modal change was of only 0·1 impulses/s.

As is evident from the examples shown in Fig. 1, complex spikes were not discharged at rigidly fixed times during the step cycle although subtle variations in the probability of discharge during different phases of the step cycle were seen when the discharges of each cell were averaged over many steps. This is evident from the

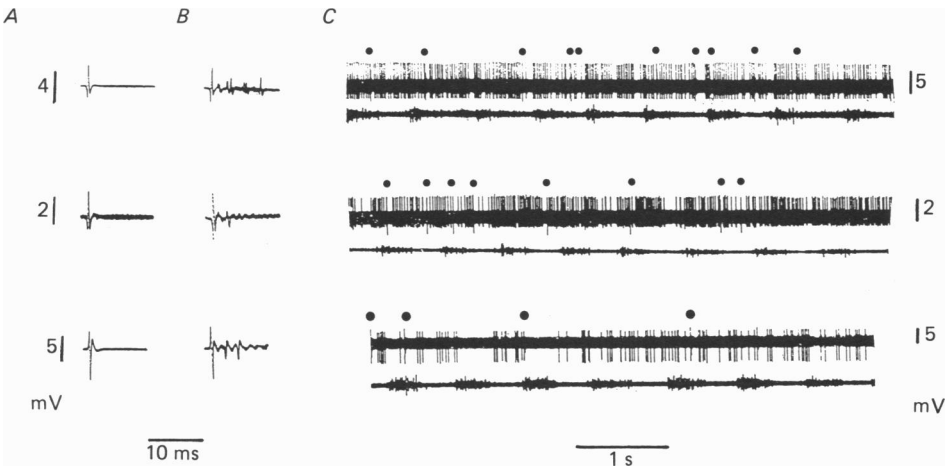

Fig. 1. The discharges of three Purkinje cells are shown. *A* and *B* show the simple and complex spikes respectively of the three cells. *C* shows the discharge during locomotion at 0·5 m/s (upper traces) together with the electromyogram of the ipsilateral triceps brachii muscle. Complex spikes are marked by dots. Negativity upwards in each case.

average rate diagrams of Fig. 2, which shows average rate diagrams for two cells with simple spike (and presumably therefore complex spike, see above) receptive fields restricted to the paw and wrist (Fig. 2*A* and *C*), and for one cell with a receptive field including the entire ipsilateral forelimb (Fig. 2*B*). However, in cases where two or more samples of the locomotor-related discharge of a single cell were available, the timings of these probability variations during the step were not consistent between samples (Fig. 2*A*–*C*; compare continuous and interrupted lines). It therefore seems probable that such variations arise by chance and do not represent a genuine step-related modulation of the complex spike discharge rates.

This possibility has been further assessed in twenty-two of the cells which were analysed using a computer and for which the discharges were recorded over sixty or more successive steps. If the discharge pattern is indeed random, the probability modulation depth (see Methods) should decrease with the square root of the sample size, i.e. equating the number of steps in the sample (n) with the sample size, a logarithmic plot of modulation depth against the number of steps should yield a straight line with slope of $-0{\cdot}5$ log Hz/log n. The modulation depth in the largest continuous sample from each of the twenty-two cells was determined from normalized post-event time histograms with a bin width of 1/10th the average step cycle duration. Such histograms were constructed for periods of ten, twenty, thirty,

etc. steps in ten-step increments up to at least sixty consecutive steps. Figure 2*D* shows the modulation depth of the three cells shown in Fig. 2*A–C* plotted as a function of the sample size in this manner. The overall average slope of such graphs for all twenty-two cells, as computed from a least-squares regression, was in fact $-0{\cdot}47 \pm 0{\cdot}15$ log Hz/log n (mean ± S.D.). Individual values from the largest samples

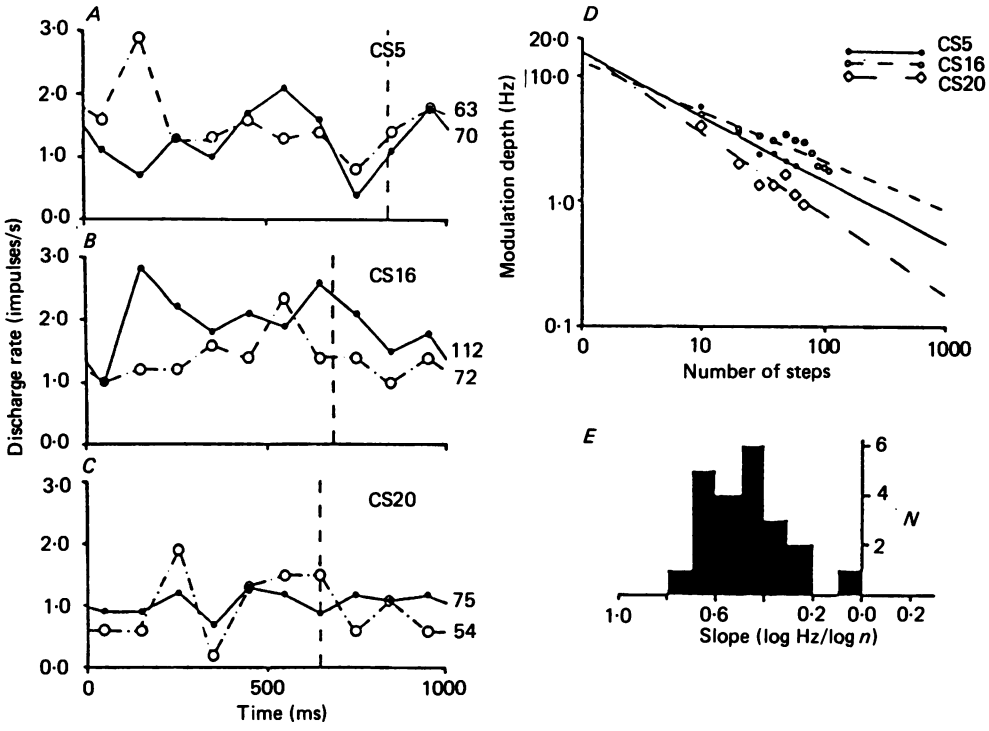

Fig. 2. *A*, *B* and *C* each show the fluctuation in the average discharge rate of the complex spikes in a Purkinje cell during stepping. Two curves are shown in each diagram, each derived from separate samples of locomotion from the same cell. The zero time reference is the onset of electromyographic activity in the ipsilateral triceps brachii and the dashed vertical lines represent the average point for the onset of the subsequent step cycle. Numbers to the right of each curve give the number of steps used in the averages. *D* shows the modulation depth as a function of sample size for the largest sample from each of the cells shown in *A–C*. Note that the axes are logarithmic. Lines drawn through the data are the least-squares regression lines. *E* shows the frequency distribution histogram of the slopes for twenty-two Purkinje cells for which graphs such as those in *D* were constructed.

for each cell ranged from $-0{\cdot}71$ to $-0{\cdot}09$ log Hz/log n and a frequency distribution histogram of the different slopes in each of the twenty-two cells is shown in Fig. 2*E*. Only two cells produced slopes that were significantly different from $-0{\cdot}5$ log Hz/log n (t test, 5% level). For the first of these two cells, three separate periods of locomotion were available and the apparent modulation pattern was not consistent between samples. Graphs of modulation depth against the number of steps were

constructed for all three samples and produced slopes of −0·46, −0·09 and +0·03 log Hz/log n with regression coefficients of 0·83, 0·40 and 0·11 respectively, i.e. the least-squares best-fit regression was for the sample with a slope of −0·46. For the second cell, only one sample of locomotion was available so that the consistency of the discharge pattern could not be assessed in this case, but the graph of modulation depth against n had a slope of −0·28 log Hz/ log n and a regression coefficient of 0·90 indicating that the data fitted a straight line well.

DISCUSSION

The present paper presents the first description of the peripheral receptive fields of paravermal Purkinje cells in awake cats. The high proportion of cells responding to tactile stimulation of the forelimb, and particularly of its distal extremities, was as expected from previous studies in anaesthetized or decerebrate cats (e.g. Thach, 1967; Ekerot & Larson, 1979) and provides confirmation that the forepaw is strongly represented in lobule V b/c.

Numerous previous investigations in the cat have shown that the discharges of complex spikes, or the spikes of inferior olive cells recorded from directly, are not frequency modulated in the course of a motor act (decerebrate locomotion: Boylls, 1980; fictitious scratching: Arshavsky, Orlovsky & Popova, 1984; active limb movements in awake cats: Gellman, Gibson & Houk, 1985; quadrupedal locomotion of awake cats: Andersson & Armstrong, 1987). These observations contrast with those of Udo *et al.* (1981) and of Kim *et al.* (1987) who have described step-related modulation in the rates of complex spike discharge during quadrupedal stepping in decerebrate cats.

The present data usefully extend the previous reports, firstly by presenting a more extensive cell sample than has previously been available from locomotor-related studies in awake cats and secondly, by presenting data from a cortical region which is among the most likely to contain Purkinje cells which respond in the manner predicted by the event-detector hypothesis of Armstrong (1974) and Rushmer *et al.* (1976). Thus, because the Purkinje cells of the present sample were recorded from a cortical area in which the predominant peripheral input arose from the paws and wrists of the forelimbs, and because complex spikes were readily evoked by light mechanical stimuli, the cells might be expected to behave in the predicted manner and discharge complex spikes either at foot touch-down or foot-lift. It may also be relevant that in respect of hindlimb-related spino-olivo-cerebellar pathways, Andersson & Sjölund (1978) have shown that those coursing in the ventral funiculus of the spinal cord and innervating the paravermal c_1 and c_3 zones of the cortex are more strongly related to peripheral events than those innervating the vermal a or b zones, which instead appear to transmit information concerning the condition of segmental interneuronal pools.

However, despite the fact that the present sample includes cells which are among the most likely to discharge complex spikes in relation to specific peripheral events, the predicted rigid phase-locking of the discharge of complex spikes to the events of the step cycle was never observed. Furthermore, although the average rate at which complex spikes were discharged in each cell did exhibit some variation during the

step when averaged over many paces, in twenty of twenty-two tested cells this variation was indistinguishable from that which might arise at random. Moreover, for one of the remaining two cells, the available evidence suggested that the apparent modulation was not genuine as it was not consistent in pattern between successive samples of steps made under the same conditions. As a consequence, there appears to be little or no frequency modulation in the discharge of complex spikes in individual Purkinje cells during the natural course of an unperturbed step.

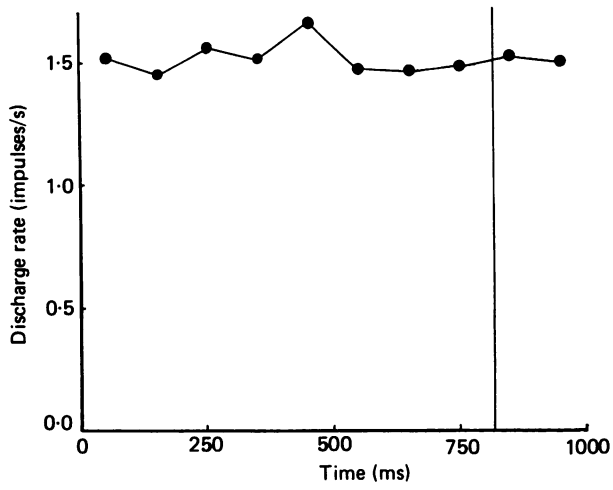

Fig. 3. The averaged complex spike discharge rate of all sixty-nine Purkinje cells. The zero time reference is the onset of electromyographic activity in the ipsilateral triceps brachii muscle and the vertical line shows the averaged point at which the next step commenced.

Gellman *et al.* (1985) and Andersson & Armstrong (1987) have recently shown that although climbing fibre activity is not affected by peripheral stimuli that result from the natural course of a movement, a large number of climbing fibres are activated when an unexpected mechanical stimulus is delivered at the periphery during a movement. Such observations can be interpreted as supporting the view of Oscarsson (1980), that the spino-olivo-cerebellar pathways act to detect errors in the course of an ongoing movement. Such a role might require a comparison of an efference copy with the peripheral feed-back and might therefore present an explanation of the results of Udo *et al.* (1981) and Kim *et al.* (1987), as in their decerebrate but actively moving cats the efference copy might be expected to be disrupted so that the postulated comparison would invariably produce an error signal. However, note that Boylls (1980) found no such modulation in the olive cells projecting to the vermal a zone even though a preparation similar to that of Udo *et al.* (1981) and Kim *et al.* (1987) was used.

Although the simultaneous activation of a large number of climbing fibres might be interpreted in the cerebellar cortex as indicating the occurrence of a movement error, it is clear from this and earlier studies that complex spikes are discharged

continually, albeit at low rate, throughout the performance of a movement. Unless these are discharged in response to small and undetected errors in the movements, individual olive cells therefore seem ill-suited to the task of indicating the occurrence of an error. What function the discharge of complex spikes might serve in the apparent absence of a motor error remains enigmatic although Boylls (1980) has suggested that the activity distributed through groups of olive cells might be capable of transmitting useful information, and that the nature of this information may not be apparent from the analysis of individual cell discharges (also see Kolb & Rubia, 1980). Figure 3 shows the result of averaging the discharge rate of complex spikes across all of the present sample of sixty-nine paravermal Purkinje cells. There is no indication here of any strongly preferred time for the occurrence of complex spikes in the population of Purkinje cells. However, the cells may have been collected from too wide an area of cortex for such modulation to be apparent even if it existed. For this reason a similar analysis has been performed using only Purkinje cells recorded from an individual cat (so that all the cells concerned were recorded from a cylinder of cortex of only *ca.* 2 mm in diameter). In two preparations there were large enough samples to make this worthwhile but no clear pattern of modulation was apparent in either case.

The present data offer the clearest refutation to date of the event-detector hypothesis presented by Armstrong (1974) and Rushmer *et al.* (1976). In addition, it has been shown that subtle variations that do occur in the discharge rate of complex spikes are not related to the peripheral events of locomotion during unperturbed stepping. In view of the highly somatotopical organization of the spino-olivo-cerebellar pathways and of the extreme sensitivity of many olive cells to peripheral tactile stimuli, these observations highlight the need for caution in interpreting the function of cerebellar afferent pathways from experiments involving peripheral stimuli imposed on the passive animal.

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